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Practical week 11

1. **Develop a multithreaded counter application using POSIX threads (pthread_create, pthread_join). Demonstrate race conditions by allowing multiple threads to update a shared variable concurrently**

Code:

```
  UW PICO 5.09                        File: 105.c                        Modified

    // Wait for all threads
    for (i = 0; i < NUM_THREADS; i++)
    {
        pthread_join(threads[i], NULL);
    }

    printf("Expected counter value: %d\n",
           NUM_THREADS * ITERATIONS);

    printf("Actual counter value:   %d\n", counter);

    if (counter == NUM_THREADS * ITERATIONS)
        printf("No race condition observed.\n");
    else
        printf("Race condition occurred!\n");

    return 0;
}

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text      ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text    ^T To Spell
```

```
nrk@nallas-mac-Book-Air ~ % nano 102.c
nrk@nallas-mac-Book-Air ~ % gcc 102.c -o 102
nrk@nallas-mac-Book-Air ~ % ./102
=== Original File ===
original.txt    Inode: 21931323  Links: 1  Size: 20 bytes

=== After Creating Hard Link ===
original.txt    Inode: 21931323  Links: 2  Size: 20 bytes
hardlink.txt    Inode: 21931323  Links: 2  Size: 20 bytes

=== After Creating Symbolic Link ===
original.txt    Inode: 21931323  Links: 2  Size: 20 bytes
hardlink.txt    Inode: 21931323  Links: 2  Size: 20 bytes
symlink.txt     Inode: 21931323  Links: 2  Size: 20 bytes

=== Link Information ===
Original and hard link have the SAME inode.
Symbolic link has a DIFFERENT inode.
symlink.txt -> original.txt
nrk@nallas-mac-Book-Air ~ %
```

2.Design a program using mmap() to perform file reading and writing. Compare memory-mapped I/O with traditional read/write operations in terms of efficiency and implementation complexity. same i have to do in ubuntu. this is practical 10.

Code:



